Fiscal Year:	FY 2012	**Task Last Updated:**	FY 11/02/2011
PI Name:	Dziegielewski, Jaroslaw Ph.D.		
Project Title:	Effects of prolonged exposure to space radiation on carcinogenesis and neuroendocrine differentiation in human prostate models		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	22908-0383	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Space Radiobiology NNJ09ZSA001N
Start Date:	01/01/2010	**End Date:**	12/31/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2013, per NSSC information (Ed., 10/10/2012) NOTE: Period of performance changed to 01/01/2010-12/31/2012 (from 11/15/09-11/14/12) per NSSC info (1/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Larner, James (University of Virginia) Parsons, Sarah (University of Virginia Health System)		
Grant/Contract No.:	NNX10AC13G		
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Performance Goal Text:			

Task Description:	The deleterious effects of radiation encountered in space presents a significant obstacle in our quest to explore the space. Unfortunately, due to the significant differences between space radiation and radiation encountered on Earth, the risks of cancer development following exposure to space radiation cannot be reliably predicted based solely on epidemiological data collected on Earth. This project will study the effects of space radiation on development and/or progression of prostate cancer. Prostate cancer is the most common cancer in men in the United States and the second leading cause of cancer deaths. It is also the most often diagnosed type of cancer in male astronauts. However, the connection between exposure to space radiation and the development of prostate cancer is not yet known. Several lines of evidence suggest that conversion of normal prostate or prostate cancer cells into cells secreting cell growth hormones, so called neuroendocrine cells, stimulate cancer development and/or progression. Strikingly, such conversion was recently reported in prostate cells exposed to prolonged irradiation. The low doses and long exposure times used in that study are not dissimilar to the doses and exposure times encountered during space travel. Therefore, we postulate that space radiation could increase the risk of prostate cancer development through the transformation of prostate cells into neuroendocrine cells during space flights. Since neuroendocrine cells seem to play an important role in the development of other important and common types of solid tumors, such as breast, lung or colon, the insight gained during this study should be also applicable to these cancers. The results obtained during the proposed studies will allow NASA to better estimate the prostate cancer health risk(s) of space exploration, as well as provide clues that will allow the development of novel preventative measures to protect astronauts.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	It is believed that space radiation can increase the risk of tumor development and progression. We hypothesize that space radiation could not only induce neoplastic transformation in prostate cells (carcinogenesis), but in addition to direct interactions with target cells (e.g. causing carcinogenic mutations in the cellular DNA), could also stimulate surrounding cells and tissues to provide growth assistance to the developing tumor. One of the mechanisms of such stimulation could be the neuroendocrine differentiation, converting normal epithelial cells into neuroendocrine-like cells which produce hormones and growth factors. Such differentiation has been observed in prostate, breast, GI tract and lung tissues, all important target organs for space radiation-induced cancers. Some of the research and technological questions that this proposal addresses include the following: 1) What is the relationship between induction of carcinogenesis and/or tumor progression and the dose and quality of space radiation? 2) Can the in vitro cell culture models be used as an approximation for prostate carcinogenesis and neuroendocrine differentiation in humans? 3) How can effective biomarkers of carcinogenic and neuroendocrine differentiation risk from space radiation be developed and validated? 4) What is the likelihood that neuroendocrine differentiation will increase the risk of carcinogenesis in prostate and other target organs? Better understanding of the mechanisms underlying the effects of exposure to radiation, including the role of stimulating activities of cancer and surrounding cells, will provide significant advantages in predicting cancer risk from space and/or terrestrial radiations.
Task Progress:	During the second year of this project, we participated in three NSRL11 campaigns, receiving two hours of proton (1000 MeV/n, LET~0.25 keV/µm), two hours of oxygen (200 MeV/n, LET~29 keV/µm), and two hours of titanium ions (1100 MeV/n, LET~106 keV/µm) irradiation time. These allowed us, together with X-ray irradiation experiments at our home institution, to establish relationship between the radiation dose and/or quality (expressed as LET) and the biological responses in irradiated prostate cells. Our results demonstrate that cellular responses to space-like ionizing radiation in prostate cells depend on radiation quality (e.g. LET) and dose, and on the cell type (normal vs. early cancer cells). LET-dependence is clearly visible in both cell survival and carcinogenic transformation, with higher LET oxygen and titanium ions causing significantly bigger effects than low LET energetic protons and X-ray photons. In addition, we have confirmed the crucial role of neurotensin signaling and neurotensin receptor 1 (NTR1) expression in prostate cancer cell survival and radioresistance. We postulate that increased NTR1 expression in irradiated normal prostate epithelial cells not only coincides with carcinogenic transformation but could be involved in the process. In the following year we will focus on dose rate-dependence, as well as determine the molecular mechanism of carcinogenic transformation in irradiated prostate cells. However, unlike carcinogenic transformation detection following exposure to single doses of radiation, we have not detected any markers (morphological changes or protein expression/secretion) of neuroendocrine differentiation in prostate cells. Therefore, we hypothesize that single-dose exposure is not capable of inducing neuroendocrine differentiation, and propose split-dose (fractionated irradiation) experiments for the next year of this project. The results obtained during this period were presented at four national and international conferences, and contributed to a peer-reviewed publication (Valerie et al., Cancer Res, 2011).
Bibliography Type:	Description: (Last Updated: 11/14/2018)
Abstracts for Journals and Proceedings	Valerie NCK, Anaganti S, DaSilva J, Parsons SJ, Larner JM, Dziegielewski J. "The effects of space radiation on carcinogenic transformation and neuroendocrine differentiation in human prostate cells." 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. , Apr-2011
Abstracts for Journals and Proceedings	Dziegielewski J, Valerie NCK, Dasilva JO, Casarez EV Jr, Anaganti S, Dunlap-Brown ME, Parsons SJ, Larner JM, Amorino GP. "Mechanisms of radiosensitization by the neurotensin receptor antagonist SR48692 in prostate cancer models." IMPaCT: Innovative Minds in Prostate Cancer Today 2011 Meeting, Orlando, FL, March 9-12, 2011. Abstract book. IMPaCT: Innovative Minds in Prostate Cancer Today 2011 Meeting, Orlando, FL, March 9-12, 2011. , Mar-2011

Abstracts for Journals and Proceedings	Dziegielewski J, Valerie NCK, Casarez EV Jr, Parsons SJ, Larner JM. "Neurotensin receptor 1 (NTR1), a novel target for tumor radiosensitization." 14th International Congress of Radiation Research, Warsaw, Poland, August 28-September 1, 2011. Abstract book. 14th International Congress of Radiation Research, Warsaw, Poland, August 28-September 1, 2011. p. 145-146. , Aug-2011
Abstracts for Journals and Proceedings	Valerie NCK, Larner JM, Parsons SJ, Dziegielewski J. "Effect of space radiation on carcinogenic transformation and neuroendocrine differentiation in human prostate cells." 22nd Annual Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011. Abstract book. 22nd Annual Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011. http://www.dsls.usra.edu/meetings/radiation2011/pdf/7057.pdf , Sep-2011
Articles in Peer-reviewed Journals	Valerie NCK, Casarez EV, DaSilva JO, Dunlap-Brown ME, Parsons SJ, Amorino GP, Dziegielewski J. "Inhibition of neurotensin receptor 1 selectively sensitizes prostate cancer to ionizing radiation." Cancer Res. 2011 Nov 1;71(21):6817-26. Epub 2011 Sep 8. PubMed PMID: 21903767 , Nov-2011